Doc1 — Product Vision & Business Requirements: HabbitQuest

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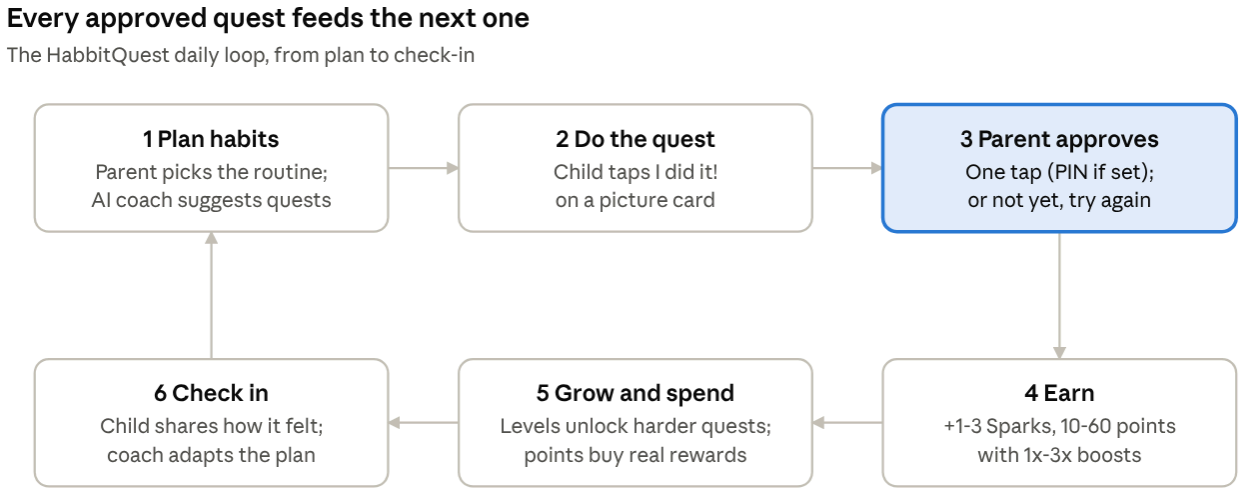
Executive summary

HabbitQuest turns a child's daily habits into picture quests that parents approve, and adds an AI habit coach that tells parents which new habits to try for the problem they describe. It serves families with children aged 3 to 13.

A child taps "I did it!", a grown-up approves (with the family PIN, if they set one), and the child earns Sparks (lifetime progress) and points (spendable currency). Sparks raise the child's level, which unlocks harder habit tiers and new worlds to explore. Points buy real rewards the parent prices, such as extra screen time or a trip to the park.

Why now: open-weight language models on low-cost inference (Groq, `gpt-oss`) make a safe, cited parenting coach affordable at consumer prices. Parents already pay for screen-time controls and chore apps, but no product combines habit routines, a reward economy, kid-safe AI and parent coaching in one place.

The working prototype is live on GitHub (rahulwarekar20/habbit-quest). It covers every v1 capability in this document, with cloud accounts, a hosted API and database, and 383 backend tests (100% coverage), 67 frontend tests and 2 browser tests that run on every change.



the HabbitQuest daily loop · 6 steps

The parent appears only twice in the loop, to plan and to approve, so the child does the work and the parent stays in control.

Problem statement

Parents know which habits they want (bed on time, homework done, less screen time) but struggle to make them stick without daily nagging. Five pain points come up repeatedly:

Pain point	What happens today	What HabbitQuest does
Nagging	Parents repeat the same reminders every day	Picture quests with clocks remind the child; the parent only approves
Not knowing what to try	Parents search forums for advice that may not fit their child's age	The habit coach recommends age-fit habits, citing guidance
Rewards that don't last	Sticker charts fade within weeks	Points, levels, streaks, golden quests, chests and surprises keep it fresh
Screen-time fights	Limits are set but switch-off causes meltdowns	Daily limits, a gentle break screen, and screen time as an earned reward
No voice for the child	Kids rarely say what feels hard	Daily check-ins with moods, stickers and safe notes

Younger children (3 to 6) cannot read lists, so most chore and habit apps built for teens or adults do not work for them. HabbitQuest is picture-first by design.

Vision, mission and principles

Vision: every child builds good habits through play, and every parent has a calm, expert coach in their pocket.

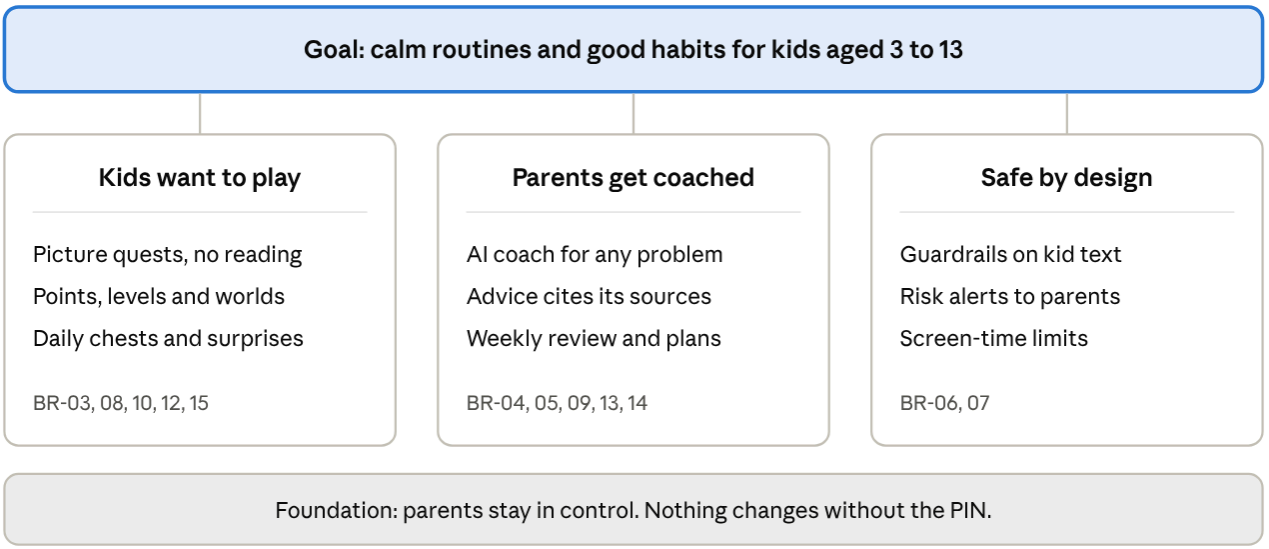
Mission: turn daily routines into quests children want to finish, and give parents trustworthy, age-fit advice on what to try next.

Product principles:

1. **Parents stay in control.** AI only suggests; every change and every reward waits for a grown-up to accept it, guarded by the family PIN when the family sets one.
2. **Pictures before words.** A 3-year-old must be able to use kid mode without reading.
3. **Never punish.** Sparks and earned points are never taken away; "not yet" invites another try.
4. **Safety first.** Everything a child reads or writes passes guardrails; signs of risk reach a parent. Every message a parent sends the coach is checked for signs a child is at risk, and gets crisis guidance instead of coaching.

5. **Small steps.** Start with easy habits and unlock harder ones as the child grows.
6. **Delight daily.** Something new or surprising appears every day, for children and parents.
7. **Private by default.** Agents see labels instead of names ("Child A"), ages and quest history only; a family's own custom habits never reach them.

Three pillars make habits stick without nagging



product pillars · goal, 3 pillars, foundation

Each pillar maps to the business requirements below; parent control underpins all three.

Target users and personas

The buyer is the parent; the daily user is the child. Each needs a different experience, on a shared device or on the child's own device signed in with the family code.

Persona	Age	Goals	Needs from HabbitQuest
Sam, busy parent	30-45	Calm routines, less nagging, fair rewards	One-tap approvals, a coach that knows their child, control over screen time
Priya, co-parent or carer	25-70	Stay consistent with the main parent	Shared view of quests, rewards and check-ins
Little explorer	3-5	Play, praise, pictures	Picture quests, a buddy, stickers instead of typing
Quest hero	6-9	Earn, collect, level up	Points, levels, chests, the reward shop, short notes

Persona	Age	Goals	Needs from HabbitQuest
Independent adventurer	10-13	Autonomy, bigger goals	Tier 3 quests, savings goals, self-starting habits

Primary market for v1: English-speaking families with at least one child aged 3 to 13 and a shared tablet or phone.

Business goals and KPIs

The first year aims to prove that families keep using HabbitQuest and that some will pay for the AI coach. The targets below are proposed starting points to validate in a pilot, not measured results.

Goal	KPI	Proposed target (year 1)
Families form the habit	Share of families active on 4+ days a week, week 4	40%
Kids engage	Quests approved per child per active day	3
Habits stick	Children with a 7-day streak in their first month	30%
Parents trust the coach	Coach recommendations accepted by parents	35%
Parents act on feedback	Check-ins read within 24 hours	70%
Safety	Unsafe AI text shown to a child	0
Revenue	Free-to-paid conversion after a 14-day trial	5%
Unit cost	AI cost per active family per month	under $0.20

Business requirements

Each requirement is a business need; Doc2 (the SRS) breaks them into testable functional requirements.

ID	Area	Requirement	Priority
BR-01	Accounts	A parent creates a family account (Google sign-in, an optional parent PIN), invites other grown-ups, and adds children aged 3 to 13	Must
BR-02	Routine	Parents choose each child's habits, times and days, from a catalog browsed by age group or from the family's own custom habits	Must
BR-03	Quests	Children mark quests done in a picture-first kid mode; parents approve or send back (with the PIN, if set)	Must
BR-04	AI coach	Parents describe a child's problem in their own words and receive recommended habits they can pick, discuss and refine	Must
BR-05	RAG	Coach advice is grounded in a curated parenting-guidance library and cites its sources	Must
BR-06	Safety	Harmful content is blocked for children aged 3 to 13; signs of risk alert the parent	Must
BR-07	Screen time	Parents set a daily kid-mode time limit and a screen-time goal; the app enforces the limit	Must
BR-08	Rewards	Children earn points with 1x, 2x and 3x boosts (golden quest, streak, double day)	Must
BR-09	Kid feedback	Children tell parents how habits feel through moods, stickers and safe notes	Must
BR-10	Levels	Levels unlock harder tiers of habits as children progress	Must
BR-11	Redemption	Children spend points on parent-priced rewards (screen time, toys, outings), approved by the parent	Must
BR-12	Surprise	Daily surprises keep kids and parents engaged (golden quest, chests, accessories, parent-planted surprises)	Must
BR-13	Insights	Parents receive a weekly AI review and an AI routine plan	Should
BR-14	Buddy	The child's buddy gives short, safe encouragement; parents can switch it off	Should

ID	Area	Requirement	Priority
BR-15	Worlds	Sparks unlock explorable worlds with hidden treasures	Could

Scope

Version 1 is the current prototype hardened for a pilot: everything in BR-01 to BR-15, with cloud accounts so a family can use several devices.

In scope for v1

- Family accounts with Google sign-in, several grown-ups, an optional parent PIN, children aged 3 to 13 with animal buddies, and kids' own devices via the family code
- A catalog of 8 goals and 41 habits in 3 tiers, browsed by age group, plus up to 30 custom habits per family
- Sparks, 6 worlds that double as levels (Starter Camp to Star Harbour), points with 1x to 3x boosts
- Reward shop, mystery box, savings goals, treasure chest, 8 buddy accessories, parent surprises
- Kid check-ins, kid-mode time limit, screen-time goal
- AI habit coach (a crew of small agents) with retrieval and citations, quest planner, weekly reviewer, buddy lines
- Word lists and a safety model on kid text and parent messages, a safety reviewer that fails closed, and parent safety alerts

Out of scope for v1

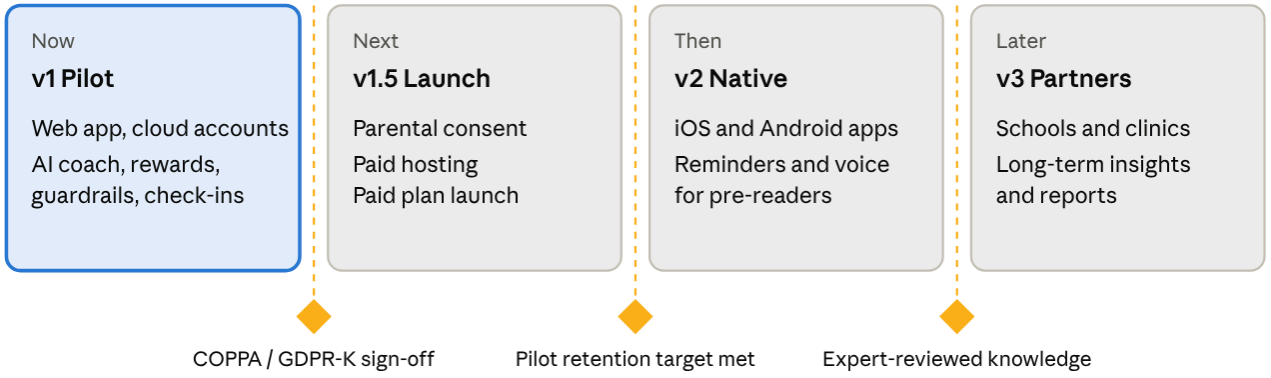
- Controlling other apps or the device's own screen time
- Real money, allowances or payments to children
- Teacher, school or clinician access
- Languages other than English

Next on the roadmap

1. Parental consent, a privacy review and paid hosting (cloud accounts, sync across devices and co-parent invites are already built)
2. Native iOS and Android apps with reminders and notifications
3. Voice for pre-readers (quests read aloud)
4. Knowledge base reviewed by child-development professionals
5. Family insights over months, and exportable reports for paediatricians

Compliance comes before growth features; cloud sync is already built

Product roadmap by phase, not to scale



roadmap · 4 phases, 3 gates

Each phase starts only when the gate before it is met; the compliance sign-off gates any public launch.

Business model

Recommended model: freemium. The quest game is free; the AI coach and family insights are a paid plan. Prices below are proposals to test, not decided.

Plan	Includes	Proposed price
Free	Routine quests, approvals, points, levels, reward shop, chests, check-ins, time limit, up to 2 children	$0
Family Plus	Everything in Free, plus the AI habit coach, weekly review, routine planner, buddy lines, unlimited children	$4.99 a month or $39.99 a year
Schools and clinics (later)	Classroom or therapy-practice dashboards	Per seat, to be decided

The paid plan carries the AI cost. Groq's open-weight models keep a family's monthly AI cost in cents, so gross margin stays high even with daily coach use.

Other revenue options to explore later: printable reward charts and merchandise, and sponsored, ad-free educational content. HabbitQuest will not show ads to children.

Assumptions, constraints, risks and compliance

The biggest launch blocker is compliance: HabbitQuest processes children's data, so it must meet children's privacy law before any public release.

Assumptions

- Families share a device or give the child their own; families who share one set a PIN and keep it private.
- Parents will pay for coaching that saves them time, if the free game proves engaging first.
- General, cited guidance is useful to parents without professional review for v1 pilots.

Constraints

- The backend, database and Google sign-in are built; the hosted database is on a free tier that expires on 28 November 2026, so production needs a paid one.
- The Groq free tier allows about 8,000 tokens a minute per model, about two coach turns a minute; production needs a paid tier.
- The app must work for children who cannot read.

Risks and mitigations

Risk	Impact	Mitigation
Unsafe AI text reaches a child	Severe	Local and AI guardrails on every kid-facing line; safe fallbacks; buddy can be switched off
Coach gives poor or harmful advice	High	Retrieval from a curated library, citations, banned-advice rules, an independent fit check and safety reviewer, professional review before launch
Children's privacy breach	Severe	Names become labels before any model call, no email or PIN to models, row-level security in the database, family export and deletion
Low retention after novelty fades	High	Daily surprises, levels and tiers, parent-planted surprises
AI provider limits or outages	Medium	Keyword fallback, Groq and Nebius presets, the game works without AI

Compliance

- **COPPA (US):** verifiable parental consent before collecting a child's data; a clear privacy notice; data deletion on request.

- **GDPR and UK GDPR, including the UK Age Appropriate Design Code:** data minimisation, high-privacy defaults, no nudges that weaken privacy, a data protection impact assessment.
- **AI providers:** name Groq as a processor in the privacy policy and sign a data processing agreement.